Data Story

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2026-10-27

Setup, Cleaning, and Testing

Setup

First things first: reading in my data!

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
library(ggplot2)
```

```
my_data <- read.csv("/Users/annarosebeckett-herbert/Desktop/quant/data_story/data/data_story_

names(my_data)
```

```
 [1] "REF_DATE"          "GEO"               "DGUID"
 [4] "Age.group.in.2019" "Gender.in.2019"    "Status"
 [7] "Indicators"        "Characteristics"   "UOM"
```

```
[10] "UOM_ID"            "SCALAR_FACTOR"     "SCALAR_ID"
[13] "VECTOR"            "COORDINATE"        "VALUE"
[16] "STATUS"            "SYMBOL"            "TERMINATED"
[19] "DECIMALS"
```

Initial Inspection

Before making any changes, I want to inspect the dataset and check the variables and categories it contains.

```
# Check the dimensions of the dataset
dim(my_data)
```

```
[1] 54648    19
```

```
# Check variable names and data types
str(my_data)
```

```
'data.frame':   54648 obs. of  19 variables:
 $ REF_DATE         : int  2023 2023 2023 2023 2023 2023 2023 2023 2023 2023 ...
 $ GEO              : chr  "Canada" "Canada" "Canada" "Canada" ...
 $ DGUID            : chr  "" "" "" "" ...
 $ Age.group.in.2019: chr  "Total, 12 to 17 years" "Total, 12 to 17 years" "Total, 12 to 17 y
 $ Gender.in.2019   : chr  "Total, Gender+" "Total, Gender+" "Total, Gender+" "Total, Gender+
 $ Status           : chr  "Reported indicator in 2019 only " "Reported indicator in 2019 onl
 $ Indicators       : chr  "Perceived health, excellent or very good" "Perceived health, exce
 $ Characteristics  : chr  "Number of persons" "Low 95% confidence interval, number of perso
 $ UOM              : chr  "Number" "Number" "Number" "Percent" ...
 $ UOM_ID           : int  223 223 223 239 239 239 223 223 223 239 ...
 $ SCALAR_FACTOR    : chr  "units" "units" "units" "units" ...
 $ SCALAR_ID        : int  0 0 0 0 0 0 0 0 0 0 ...
 $ VECTOR           : chr  "v1598161616" "v1598161617" "v1598161618" "v1598161619" ...
 $ COORDINATE       : chr  "1.1.1.1.1.1" "1.1.1.1.1.2" "1.1.1.1.1.3" "1.1.1.1.1.4" ...
 $ VALUE            : num  423500 389100 457900 18.6 17.1 ...
 $ STATUS           : chr  "" "" "" "" ...
 $ SYMBOL           : logi  NA NA NA NA NA NA ...
 $ TERMINATED       : logi  NA NA NA NA NA NA ...
 $ DECIMALS         : int  0 0 0 1 1 1 0 0 0 1 ...
```

```r
# Check the categories in the main variables
unique(my_data$GEO)
```

```
 [1] "Canada"                    "Newfoundland and Labrador"
 [3] "Prince Edward Island"      "Nova Scotia"
 [5] "New Brunswick"             "Quebec"
 [7] "Ontario"                   "Manitoba"
 [9] "Saskatchewan"              "Alberta"
[11] "British Columbia"
```

```r
unique(my_data$Indicators)
```

```
 [1] "Perceived health, excellent or very good"
 [2] "Perceived health, fair or poor"
 [3] "Perceived mental health, excellent or very good"
 [4] "Perceived mental health, fair or poor"
 [5] "Smoking, daily or occasionally"
 [6] "Smoking, daily"
 [7] "Cannabis use in the past 12 months"
 [8] "Cannabis use in the past 12 months, daily or almost daily"
 [9] "Used  e-cigarette or vaping device, past 30 days"
[10] "Self-reported physical activity, average 60 or more minutes per day"
[11] "Headache"
[12] "Stomach ache"
[13] "Backache"
[14] "Difficulties in getting to sleep"
[15] "Changed eating habits to manage weight"
[16] "Changed eating habits to manage weight, daily"
[17] "Life satisfaction, satisfied or very satisfied"
[18] "Met sleep guidelines in the last 7 days"
[19] "Preoccupied with a desire to be thinner"
[20] "Preoccupied with a desire to be thinner, daily"
[21] "Vomited to lose weight"
[22] "Vomited to lose weight, daily"
[23] "Perceived life stress, most days quite a bit or extremely stressful"
```

```r
unique(my_data$`Age group in 2019`)
```

```
NULL
```

```
unique(my_data$`Gender in 2019`)
```

```
NULL
```

```
unique(my_data$Status)
```

```
[1] "Reported indicator in 2019 only "
[2] "Reported indicator in 2023 only"
[3] "Reported indicator in 2019 and 2023"
[4] "Did not report indicator in  2019 and 2023"
```

```
unique(my_data$Characteristics)
```

```
[1] "Number of persons"
[2] "Low 95% confidence interval, number of persons"
[3] "High 95% confidence interval, number of persons"
[4] "Percent"
[5] "Low 95% confidence interval, percent"
[6] "High 95% confidence interval, percent"
```

```
# Check missing values
colSums(is.na(my_data))
```

```
        REF_DATE              GEO             DGUID Age.group.in.2019
               0                0                 0                 0
  Gender.in.2019           Status       Indicators   Characteristics
               0                0                 0                 0
             UOM           UOM_ID     SCALAR_FACTOR         SCALAR_ID
               0                0                 0                 0
          VECTOR       COORDINATE             VALUE            STATUS
               0                0             23940                 0
          SYMBOL       TERMINATED          DECIMALS
           54648            54648                 0
```

Selecting and Cleaning Relevant Data

First, I am selecting just my indicator of interest, which is perceived mental health rated as excellent or very good. I am also limiting the data to just Canada rather than including provincial estimates.

```r
my_data <- my_data |>
  filter(
    GEO == "Canada",
    Indicators == "Perceived mental health, excellent or very good"
  )

head(my_data)
```

```
  REF_DATE    GEO DGUID     Age.group.in.2019 Gender.in.2019
1     2023 Canada       Total, 12 to 17 years Total, Gender+
2     2023 Canada       Total, 12 to 17 years Total, Gender+
3     2023 Canada       Total, 12 to 17 years Total, Gender+
4     2023 Canada       Total, 12 to 17 years Total, Gender+
5     2023 Canada       Total, 12 to 17 years Total, Gender+
6     2023 Canada       Total, 12 to 17 years Total, Gender+
                           Status
1 Reported indicator in 2019 only
2 Reported indicator in 2019 only
3 Reported indicator in 2019 only
4 Reported indicator in 2019 only
5 Reported indicator in 2019 only
6 Reported indicator in 2019 only
                                      Indicators
1 Perceived mental health, excellent or very good
2 Perceived mental health, excellent or very good
3 Perceived mental health, excellent or very good
4 Perceived mental health, excellent or very good
5 Perceived mental health, excellent or very good
6 Perceived mental health, excellent or very good
                                  Characteristics     UOM UOM_ID SCALAR_FACTOR
1                                Number of persons  Number    223         units
2  Low 95% confidence interval, number of persons  Number    223         units
3 High 95% confidence interval, number of persons  Number    223         units
4                                        Percent Percent    239         units
5            Low 95% confidence interval, percent Percent    239         units
6           High 95% confidence interval, percent Percent    239         units
  SCALAR_ID      VECTOR  COORDINATE    VALUE STATUS SYMBOL TERMINATED DECIMALS
1         0 v1598161628 1.1.1.1.3.1 705000.0             NA         NA        0
2         0 v1598161629 1.1.1.1.3.2 664800.0             NA         NA        0
3         0 v1598161630 1.1.1.1.3.3 745100.0             NA         NA        0
4         0 v1598161631 1.1.1.1.3.4     31.2             NA         NA        1
5         0 v1598161632 1.1.1.1.3.5     29.4             NA         NA        1
```

```
6         0 v1598161633 1.1.1.1.3.6     33.0            NA          NA        1
```

Renaming the column names next, for clarity:

```
names(my_data)[names(my_data) == "Age.group.in.2019"] <- "Age"
names(my_data)[names(my_data) == "Gender.in.2019"] <- "Gender"
names(my_data)[names(my_data) == "Indicators"] <- "Indicator"
names(my_data)[names(my_data) == "Characteristics"] <- "MetricType"
names(my_data)[names(my_data) == "VALUE"] <- "Value"

names(my_data)
```

```
 [1] "REF_DATE"      "GEO"           "DGUID"         "Age"
 [5] "Gender"        "Status"        "Indicator"     "MetricType"
 [9] "UOM"           "UOM_ID"        "SCALAR_FACTOR" "SCALAR_ID"
[13] "VECTOR"        "COORDINATE"    "Value"         "STATUS"
[17] "SYMBOL"        "TERMINATED"    "DECIMALS"
```

Now I want to remove the "total" categories for both age and gender, since I am interested in comparing the two age groups and two gender categories separately:

```
my_data <- my_data |>
  filter(
    Age != "Total, 12 to 17 years",
    Gender != "Total, Gender+"
  )

head(my_data)
```

```
  REF_DATE    GEO DGUID            Age Gender                          Status
1     2023 Canada       12 to 14 years  Boys+ Reported indicator in 2019 only
2     2023 Canada       12 to 14 years  Boys+ Reported indicator in 2019 only
3     2023 Canada       12 to 14 years  Boys+ Reported indicator in 2019 only
4     2023 Canada       12 to 14 years  Boys+ Reported indicator in 2019 only
5     2023 Canada       12 to 14 years  Boys+ Reported indicator in 2019 only
6     2023 Canada       12 to 14 years  Boys+ Reported indicator in 2019 only
                                      Indicator
1 Perceived mental health, excellent or very good
2 Perceived mental health, excellent or very good
3 Perceived mental health, excellent or very good
4 Perceived mental health, excellent or very good
```

```
5 Perceived mental health, excellent or very good
6 Perceived mental health, excellent or very good
                                      MetricType     UOM UOM_ID SCALAR_FACTOR
1                                Number of persons  Number    223         units
2  Low 95% confidence interval, number of persons  Number    223         units
3 High 95% confidence interval, number of persons  Number    223         units
4                                        Percent Percent    239         units
5            Low 95% confidence interval, percent Percent    239         units
6           High 95% confidence interval, percent Percent    239         units
  SCALAR_ID      VECTOR  COORDINATE    Value STATUS SYMBOL TERMINATED DECIMALS
1         0 v1598163836 1.2.2.1.3.1 178700.0            NA         NA        0
2         0 v1598163837 1.2.2.1.3.2 155600.0            NA         NA        0
3         0 v1598163838 1.2.2.1.3.3 201800.0            NA         NA        0
4         0 v1598163839 1.2.2.1.3.4     29.7            NA         NA        1
5         0 v1598163840 1.2.2.1.3.5     26.1            NA         NA        1
6         0 v1598163841 1.2.2.1.3.6     33.2            NA         NA        1
```

I am also selecting only the percentage estimates, rather than the number of persons or confidence interval estimates.

```
my_data <- my_data |>
  filter(MetricType == "Percent")

head(my_data)
```

```
  REF_DATE    GEO DGUID            Age Gender
1     2023 Canada       12 to 14 years  Boys+
2     2023 Canada       12 to 14 years  Boys+
3     2023 Canada       12 to 14 years  Boys+
4     2023 Canada       12 to 14 years  Boys+
5     2023 Canada       12 to 14 years Girls+
6     2023 Canada       12 to 14 years Girls+
                                    Status
1           Reported indicator in 2019 only
2            Reported indicator in 2023 only
3        Reported indicator in 2019 and 2023
4 Did not report indicator in  2019 and 2023
5           Reported indicator in 2019 only
6            Reported indicator in 2023 only
                                      Indicator MetricType     UOM UOM_ID
1 Perceived mental health, excellent or very good    Percent Percent    239
2 Perceived mental health, excellent or very good    Percent Percent    239
```

```
3 Perceived mental health, excellent or very good    Percent Percent    239
4 Perceived mental health, excellent or very good    Percent Percent    239
5 Perceived mental health, excellent or very good    Percent Percent    239
6 Perceived mental health, excellent or very good    Percent Percent    239
  SCALAR_FACTOR SCALAR_ID      VECTOR  COORDINATE Value STATUS SYMBOL
1         units         0 v1598163839 1.2.2.1.3.4  29.7            NA
2         units         0 v1598163977 1.2.2.2.3.4   9.5            NA
3         units         0 v1598164115 1.2.2.3.3.4  50.2            NA
4         units         0 v1598164253 1.2.2.4.3.4  10.6            NA
5         units         0 v1598164391 1.2.3.1.3.4  38.7            NA
6         units         0 v1598164529 1.2.3.2.3.4   5.9      E     NA
  TERMINATED DECIMALS
1         NA        1
2         NA        1
3         NA        1
4         NA        1
5         NA        1
6         NA        1
```

Checking variable types and values

Now I want to check that the value column is numeric, since I will be using it to calculate percentages and changes.

```
# Check variable types
str(my_data)
```

```
'data.frame':   16 obs. of  19 variables:
 $ REF_DATE     : int  2023 2023 2023 2023 2023 2023 2023 2023 2023 2023 ...
 $ GEO          : chr  "Canada" "Canada" "Canada" "Canada" ...
 $ DGUID        : chr  "" "" "" "" ...
 $ Age          : chr  "12 to 14 years" "12 to 14 years" "12 to 14 years" "12 to 14 years" .
 $ Gender       : chr  "Boys+" "Boys+" "Boys+" "Boys+" ...
 $ Status       : chr  "Reported indicator in 2019 only " "Reported indicator in 2023 only" "
 $ Indicator    : chr  "Perceived mental health, excellent or very good" "Perceived mental he
 $ MetricType   : chr  "Percent" "Percent" "Percent" "Percent" ...
 $ UOM          : chr  "Percent" "Percent" "Percent" "Percent" ...
 $ UOM_ID       : int  239 239 239 239 239 239 239 239 239 239 ...
 $ SCALAR_FACTOR: chr  "units" "units" "units" "units" ...
 $ SCALAR_ID    : int  0 0 0 0 0 0 0 0 0 0 ...
 $ VECTOR       : chr  "v1598163839" "v1598163977" "v1598164115" "v1598164253" ...
```

```
 $ COORDINATE   : chr  "1.2.2.1.3.4" "1.2.2.2.3.4" "1.2.2.3.3.4" "1.2.2.4.3.4" ...
 $ Value        : num  29.7 9.5 50.2 10.6 38.7 5.9 31.3 24 27.6 9.2 ...
 $ STATUS       : chr  "" "" "" "" ...
 $ SYMBOL       : logi  NA NA NA NA NA NA ...
 $ TERMINATED   : logi  NA NA NA NA NA NA ...
 $ DECIMALS     : int  1 1 1 1 1 1 1 1 1 1 ...
```

```r
# Check that Value is numeric
is.numeric(my_data$Value)
```

```
[1] TRUE
```

```r
# Check the remaining categories
unique(my_data$Age)
```

```
[1] "12 to 14 years" "15 to 17 years"
```

```r
unique(my_data$Gender)
```

```
[1] "Boys+"  "Girls+"
```

```r
unique(my_data$Status)
```

```
[1] "Reported indicator in 2019 only "
[2] "Reported indicator in 2023 only"
[3] "Reported indicator in 2019 and 2023"
[4] "Did not report indicator in  2019 and 2023"
```

Looks good!

Combining categories and calculating change

Since I am interested in comparing 2019 and 2023, I need to combine some of the reporting categories. “Reported indicator in 2019 only” and “Reported indicator in 2019 and 2023” will be combined to calculate the percentage reporting excellent or very good mental health in 2019. Similarly, “Reported indicator in 2023 only” and “Reported indicator in 2019 and 2023” will be combined to calculate the percentage for 2023.

First, I will simplify the status labels and standardize them by removing inconsistent spacing in the original dataset.

```
my_data$Status <- stringr::str_squish(my_data$Status)

my_data <- my_data |>
  mutate(
    Status = case_when(
      Status == "Reported indicator in 2019 only" ~ "2019_only",
      Status == "Reported indicator in 2023 only" ~ "2023_only",
      Status == "Reported indicator in 2019 and 2023" ~ "both",
      Status == "Did not report indicator in 2019 and 2023" ~ "neither"
    )
  )

unique(my_data$Status)
```

```
[1] "2019_only" "2023_only" "both"      "neither"
```

Now, reshaping the data so each status becomes its own column:

```
reshaped_data <- my_data |>
  select(Age, Gender, Status, Value) |>
  pivot_wider(
    names_from = Status,
    values_from = Value
  )

reshaped_data
```

```
# A tibble: 4 x 6
  Age            Gender `2019_only` `2023_only`  both neither
  <chr>          <chr>        <dbl>       <dbl> <dbl>   <dbl>
1 12 to 14 years Boys+         29.7         9.5  50.2    10.6
2 12 to 14 years Girls+        38.7         5.9  31.3    24
3 15 to 17 years Boys+         27.6         9.2  39      24.2
4 15 to 17 years Girls+        28.4         9.1  20.4    42.2
```

And calculating the percents for both years, plus the change:

```
reshaped_data <- reshaped_data |>
  mutate(
    percent_2019 = `2019_only` + both,
```

```
    percent_2023 = `2023_only` + both,
    change = percent_2023 - percent_2019
  )

reshaped_data
```

```
# A tibble: 4 x 9
  Age     Gender `2019_only` `2023_only`  both neither percent_2019 percent_2023
  <chr>   <chr>        <dbl>       <dbl> <dbl>   <dbl>        <dbl>        <dbl>
1 12 to ~ Boys+         29.7         9.5  50.2    10.6         79.9         59.7
2 12 to ~ Girls+        38.7         5.9  31.3    24           70           37.2
3 15 to ~ Boys+         27.6         9.2  39      24.2         66.6         48.2
4 15 to ~ Girls+        28.4         9.1  20.4    42.2         48.8         29.5
# i 1 more variable: change <dbl>
```

Testing

First, I will check that the categories are complete and that the filtering process has not inadvertently excluded observations by seeing if the four categories add up to approximately 100%:

```
status_totals <- my_data |>
  group_by(Age, Gender) |>
  summarise(
    total_percent = sum(Value),
    .groups = "drop"
  )

status_totals
```

```
# A tibble: 4 x 3
  Age            Gender total_percent
  <chr>          <chr>          <dbl>
1 12 to 14 years Boys+          100
2 12 to 14 years Girls+          99.9
3 15 to 17 years Boys+          100
4 15 to 17 years Girls+         100.
```

```
all(abs(status_totals$total_percent - 100) < 0.2)
```

```
[1] TRUE
```

I will also now check for any missing or invalid values.

```
all(!is.na(my_data$Value))
```

```
[1] TRUE
```

```
all(my_data$Value >= 0 & my_data$Value <= 100)
```

```
[1] TRUE
```

Lastly, I want to check that my addition of the two years was correct and that the calculated change is consistent with the two yearly values:

```
all(reshaped_data$percent_2019 >= 0 &
      reshaped_data$percent_2019 <= 100)
```

```
[1] TRUE
```

```
all(reshaped_data$percent_2023 >= 0 &
      reshaped_data$percent_2023 <= 100)
```

```
[1] TRUE
```

```
all(
  abs(
    reshaped_data$change -
      (reshaped_data$percent_2023 - reshaped_data$percent_2019)
  ) < 0.0001
)
```

```
[1] TRUE
```

Good to go!

Exploratory Analysis

First, I will reshape my data into a format that is more suitable for plotting:

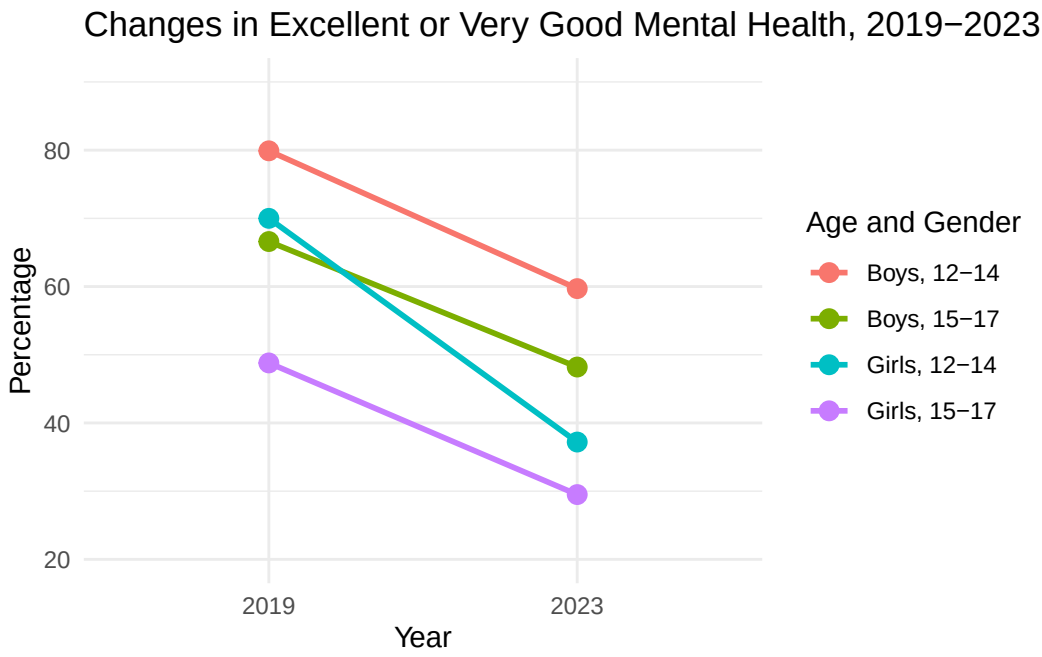
```
plot_data <- reshaped_data |>
  select(Age, Gender, percent_2019, percent_2023) |>
  pivot_longer(
    cols = c(percent_2019, percent_2023),
    names_to = "Year",
    values_to = "Percentage"
  ) |>
  mutate(
    Year = recode(
      Year,
      percent_2019 = "2019",
      percent_2023 = "2023"
    )
  )
```

Now let's make a graph comparing 2019 and 2023!

```
plot_data <- plot_data |>
  mutate(
    Group = case_when(
      Age == "12 to 14 years" & Gender == "Boys+" ~ "Boys, 12-14",
      Age == "15 to 17 years" & Gender == "Boys+" ~ "Boys, 15-17",
      Age == "12 to 14 years" & Gender == "Girls+" ~ "Girls, 12-14",
      Age == "15 to 17 years" & Gender == "Girls+" ~ "Girls, 15-17"
    )
  )

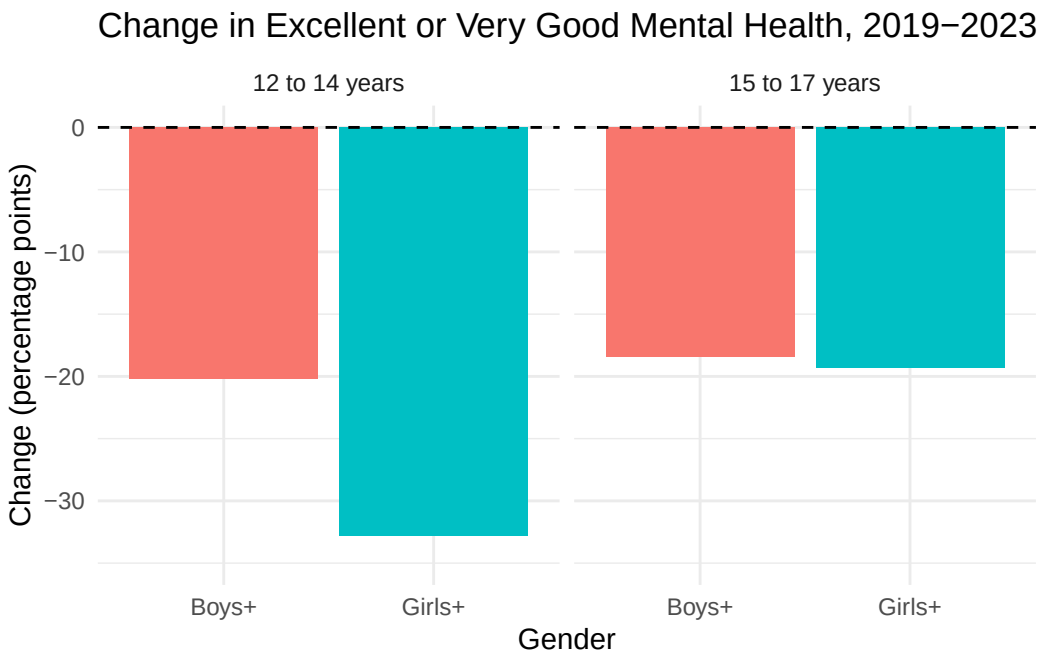
ggplot(plot_data, aes(
  x = Year,
  y = Percentage,
  color = Group,
  group = Group
)) +
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
  ylim(20,90) +
  labs(
    title = "Changes in Excellent or Very Good Mental Health, 2019-2023",
```

```
    x = "Year",
    y = "Percentage",
    color = "Age and Gender"
  ) +
  theme_minimal()
```

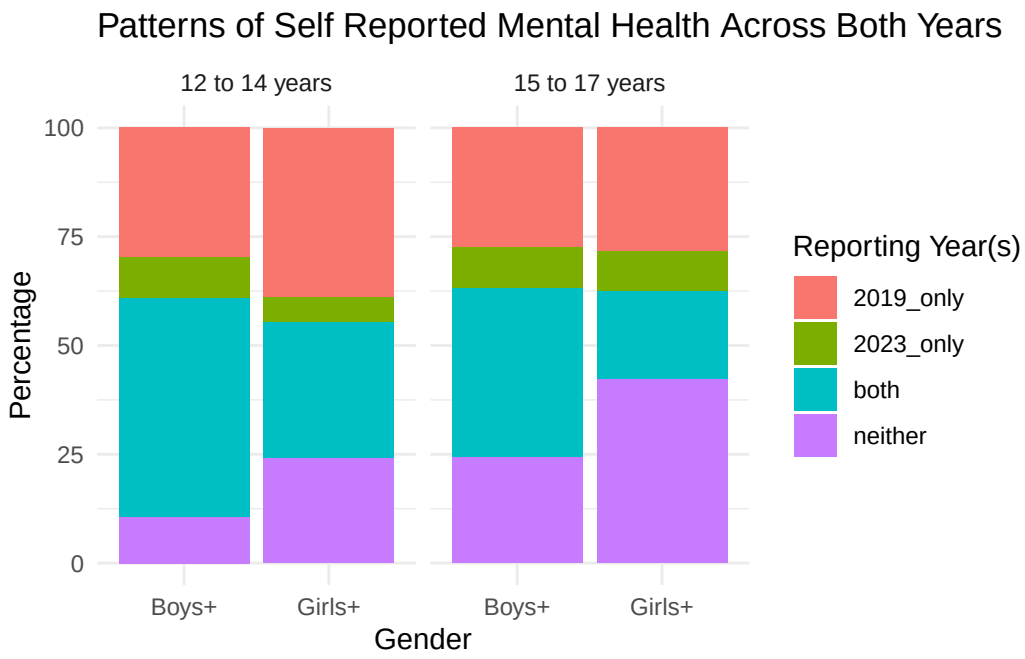


And as a percent change:

```
ggplot(reshaped_data, aes(x = Gender, y = change, fill = Gender)) +
  geom_col() +
  facet_wrap(~ Age) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  ylim(-35,0) +
  labs(
    title = "Change in Excellent or Very Good Mental Health, 2019-2023",
    x = "Gender",
    y = "Change (percentage points)"
  ) +
  theme_minimal() +
  guides(fill = "none")
```

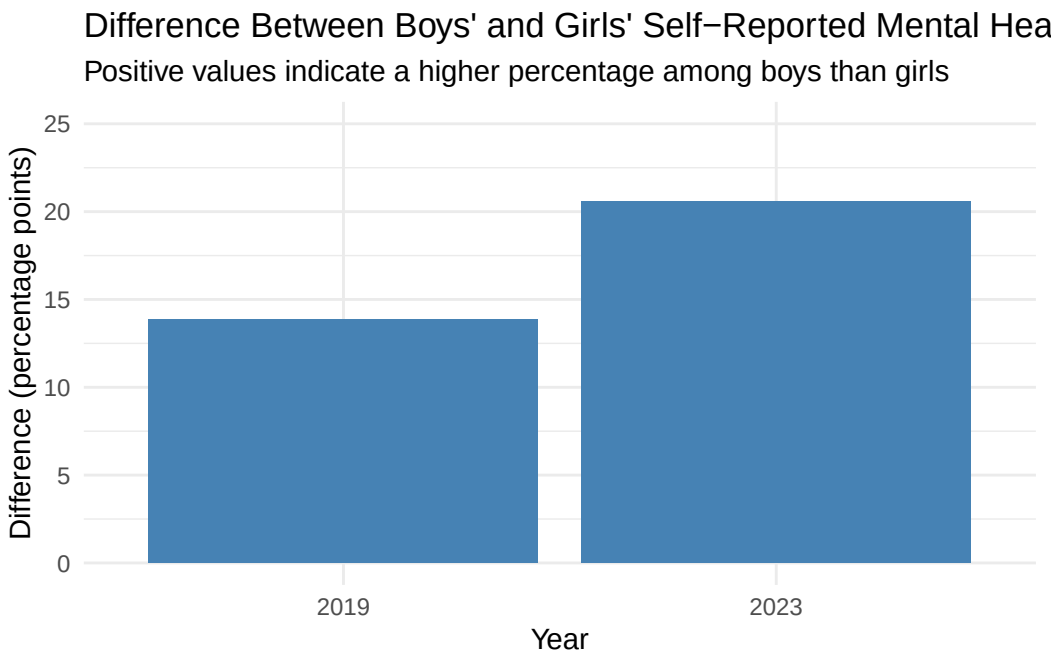



```
ggplot(my_data, aes(x = Gender, y = Value, fill = Status)) +
  geom_col() +
  facet_wrap(~ Age) +
  labs(
    title = "Patterns of Self Reported Mental Health Across Both Years",
    x = "Gender",
    y = "Percentage",
    fill = "Reporting Year(s)"
  ) +
  theme_minimal()
```




```r
gap_data <- data.frame(
  Year = c("2019", "2023"),
  GenderGap = c(13.85, 20.6)
)

ggplot(gap_data, aes(x = Year, y = GenderGap)) +
  geom_col(fill = "steelblue") +
  ylim(0,25)+
  labs(
    title = "Difference Between Boys' and Girls' Self-Reported Mental Health",
    subtitle = "Positive values indicate a higher percentage among boys than girls",
    x = "Year",
    y = "Difference (percentage points)"
  ) +
  theme_minimal()
```



Surprises, Takeaways, etc.